

	1	2	3	4	5	6	7	8
	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
candidate blocks ($a:b$)					$e_{n,k}(a,b)$			
					$k=1$	$k=2$	$k=3$	
[1:2]					-1.20	-0.30	-2.10	
[3:5]					-0.70	-1.10	-0.40	
[5:6]					-1.50	-0.20	-1.00	
[6:8]					-0.60	-0.80	-0.10	

$$\log Z_n(U) = \log \sum_{S_n, z_n} p(S_n, z_n) \prod_{\ell} e_{n, z_\ell}(a_\ell, b_\ell) \text{ all legal labelled segmentations of trace } n$$

(accumulate total path mass)
 $\alpha_n(j, k)$: total mass of segmentations ending at position j with skill k

Legend:

- solid: append block $(a:b)$ with weight $e_{n,k}(a,b)$
- dashed: only arcs ending at j enter $\alpha_n(j, k)$
- coloured: the worked example of the main figure

backward FFBS sampling of one complete
labelled segmentation (S_n, z_n)

$z_{n,1}=2$ $z_{n,2}=1$ $z_{n,3}=3$

 $(S_n, z_n) \sim p(S_n, z_n \mid \mathbf{x}^{(n)}, U, \vartheta)$

```

graph TD
    A["current  $U$ ,  $H = h(U)$ "] --> B[" $\log Z_1(U)$ "]
    A --> C[" $\log Z_2(U)$ "]
    A --> D[" $\dots \log Z_N(U)$ "]
    B --> E[" $\sum_n \log Z_n(U)$ "]
    C --> E
    D --> E
    E --> F["propose  $U'$ ,  $H' = h(U')$ "]
    F --> G["compute  $\sum_n \log Z_n(U')$   
by the forward DP"]
    G --> H["accept with MH probability"]
    H -- yes --> I["set  $U \leftarrow U'$ ;  
refresh  $(S_n, z_n)$  for  
all  $n$  by FFBS"]
    H -- no --> J["keep  $U$ ;  
current  $(S_n, z_n)$   
remain valid"]

```

U' is scored against all segmentations, not only the current sam

one forward DP, two uses: (1) backward-sample latent paths (S_n, z_n) via FFBS; (2) path-marginal scores $\log Z_n(U)$ for structure updates